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Analyzing a VMware Environment

Risk and duration: Migration to OpenShift Virtualization ·

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Agenda

- Capture the VMware estate in a structured way
- Assess risks for OpenShift Virtualization
- Estimate duration and phases realistically
- Use MTV as the Red Hat migration tool
- Conclusion and recommended next steps

The Cost of the Dependency Blind Spot

73% of failed migration projects discover **critical dependencies** after the migration has already started.

Only 19% of projects succeed smoothly—and they all **complete comprehensive dependency mapping** before lift-off.

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Phase 1: Translating the Architectural Language

vSphere Datastore	Persistent Volume (PV/PVC)
vMotion	Live Migration
vSwitch / DvSwitch	NMState / Multus
DRS (Dynamic Resource Scheduling)	Pod Eviction & Descheduler

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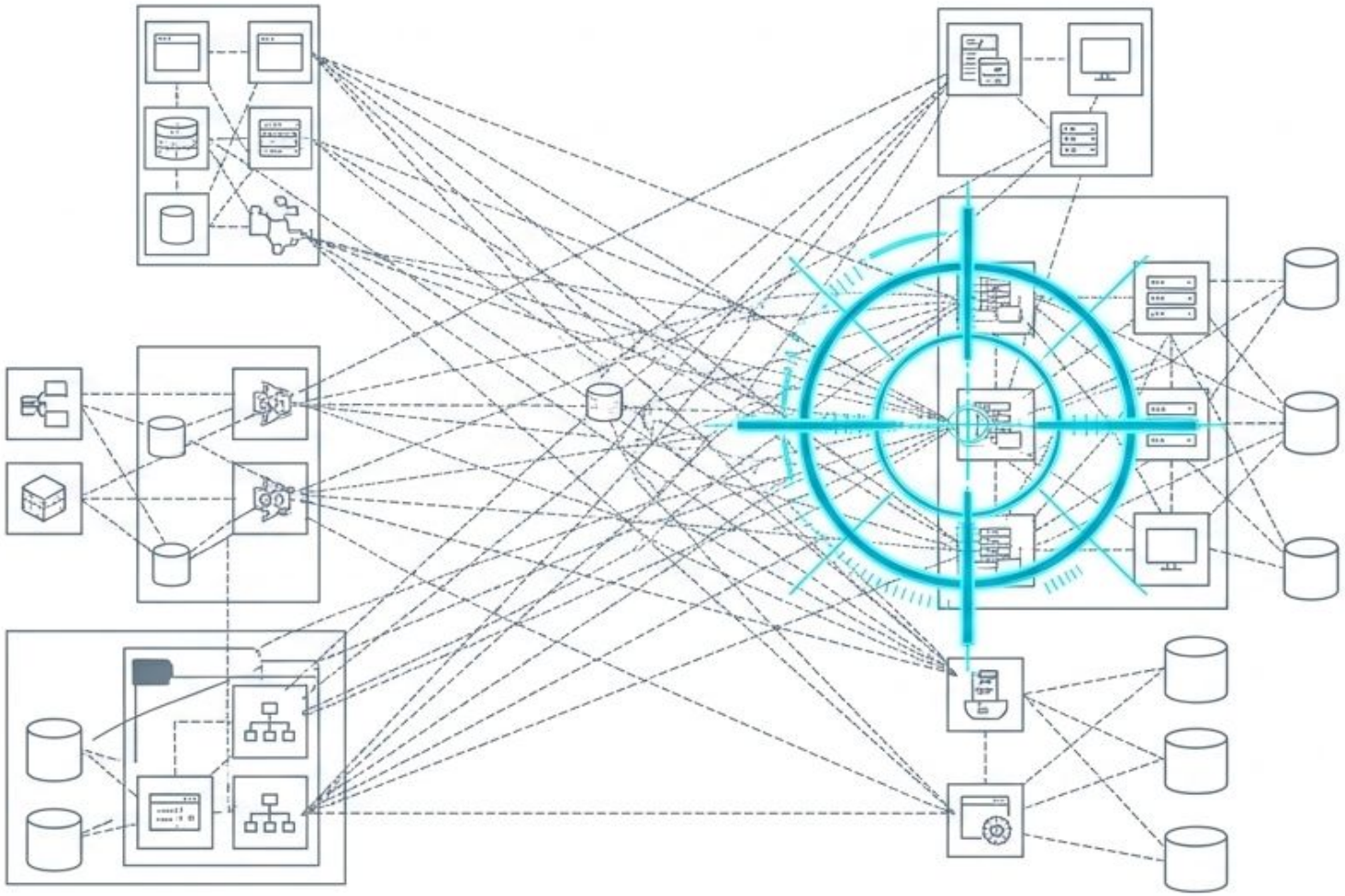
Mapping True Application Boundaries



Stop Guessing. Start Verifying.

- **Automated Flow Analysis:** Utilize tools like vRealize Network Insight to capture actual traffic.
- **Policy Transformation:** Replace thousands of lines of undocumented physical firewall rules with modern, flow-based NSX-T or container network policies.
- Identify multi-cloud fragmentation risks.

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The Ultimate Metric: Data Gravity

[Volume Transferred (TB)] × [Network Latency (ms)] = Data Gravity Score

Diagnostic Example:
An application reads **2 TB** of data at **30ms** latency = 60,000 TBxms.
If the app moves to the cloud but the database stays on-prem, and latency jumps to **80ms**, performance will **catastrophically collapse**.

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Phase 2: Defining Your Environment Typology

Type 1: The Agnostic Environment

- Provisioned via infrastructure-agnostic automation (Terraform, Ansible).
- Minimal proprietary VMware integrations.

Type 2: The Deeply Integrated Environment

- Deeply woven into VMware's Software Defined Data Center.
- Heavy reliance on NSX for networking, vSAN for storage, Site Recovery Manager for DR.

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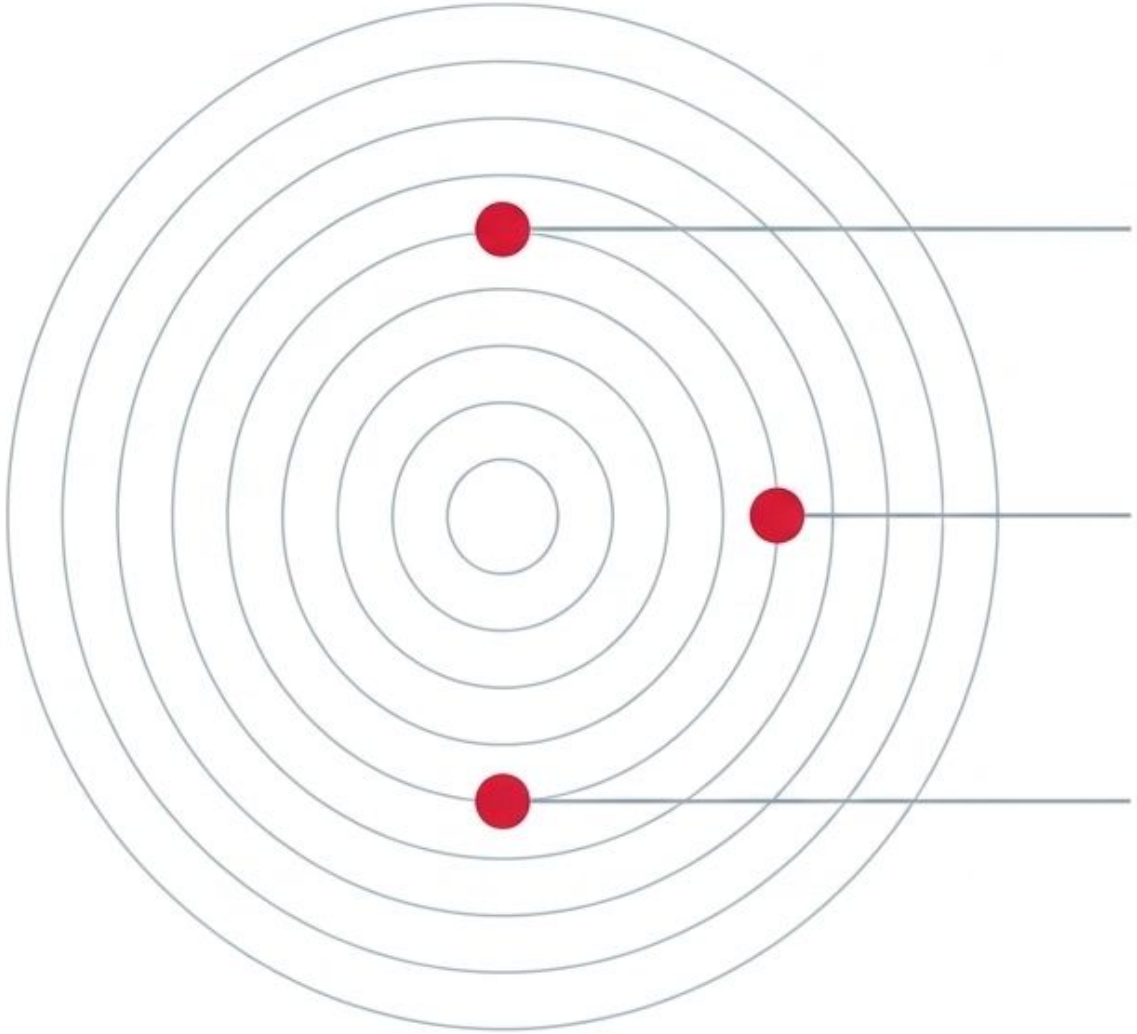
Navigating the Migration Torpedoes



The 3 Silent Killers:

1. Undocumented Active Directory / LDAP Trusts
Brittle, legacy domain trusts that have no clean equivalent in cloud-native Identity Providers

2. Hard-Coded IPs & Secrets
Configurations that bypass DNS and will break the instant the network boundary shifts

3. Single-Threaded File Locks
Applications demanding zero-downtime cutovers but constrained by legacy file-locking architectures

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Phase 3: The 2-Week Rapid Discovery Framework

Days 1–3: Connect & Collect

Establish vCenter API access; ingest raw inventory automatically.

Days 4–6: Normalize & Categorize

Standardize naming; identify quick wins vs. blockers.

Days 7–9: Analyze & Quantify

Calculate total scope, apply benchmarks, estimate wave capacity.

Days 10–14: Synthesize

Produce the executive risk register and actionable next steps.

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Phase 4: Execution via Red Hat MTV

Red Hat Migration Toolkit for Virtualization (MTV)

- At-scale, automated migration.
- Supports bulk orchestration of both warm and cold migrations.
- Translates legacy VM configurations into Kubernetes-native VirtualMachine instances.

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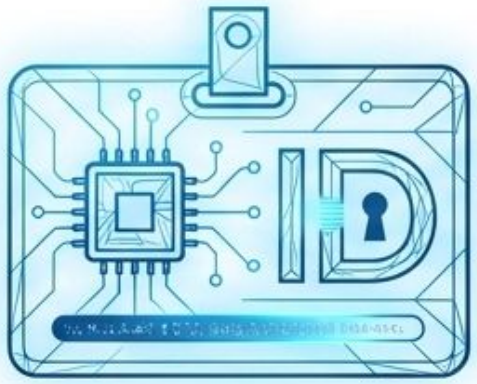
Securing the Migration Pipeline

Configure Root CA

Establish explicit, encrypted trust for all API communication between OpenShift and vCenter providers.



Long-Lived Service Accounts

Migrations execute in continuous waves over months. Relying on temporary user tokens causes pipeline failures mid-flight. **Persistent permissions** are mandatory.

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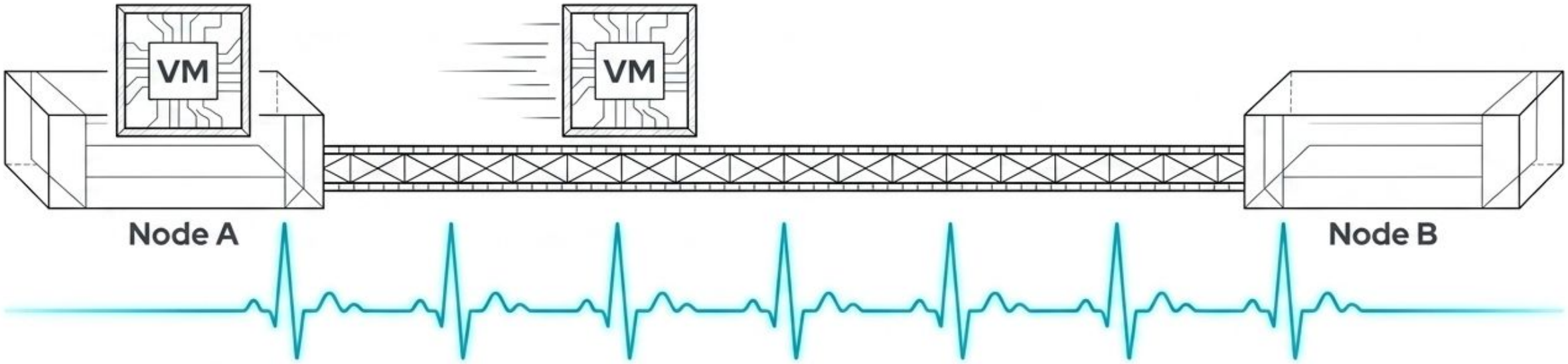
Achieving Cross-Cluster Live Migration



Zero-Downtime Execution

- The OpenShift equivalent of vMotion, moving running VMs between nodes without interruption.
- Prerequisite: Target cluster shared storage configured with **ReadWriteMany (RWX)** access.
- Prerequisite: Sufficient spare RAM capacity to support node drains.
- All live migration traffic is automatically **TLS-encrypted**.

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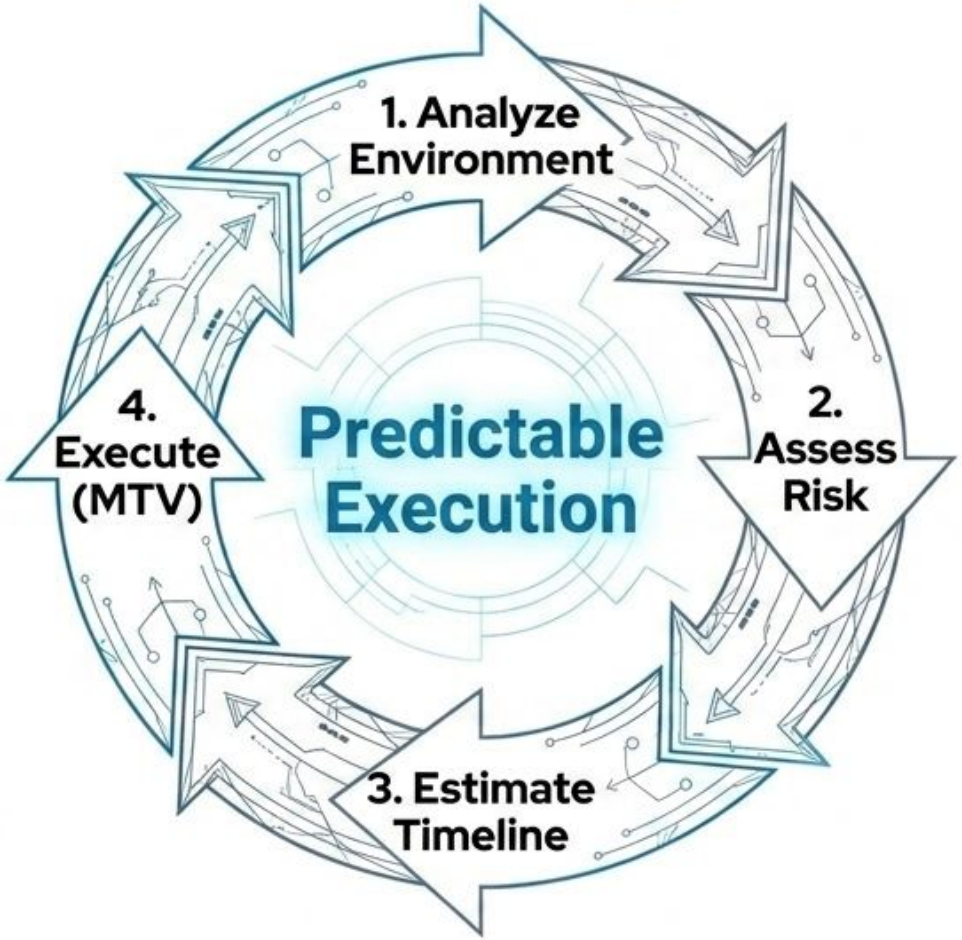
The Predictable Migration Engine



From blind spots to precision execution.
Great migrations do not start with blueprints; they start with high-fidelity maps of the risks.

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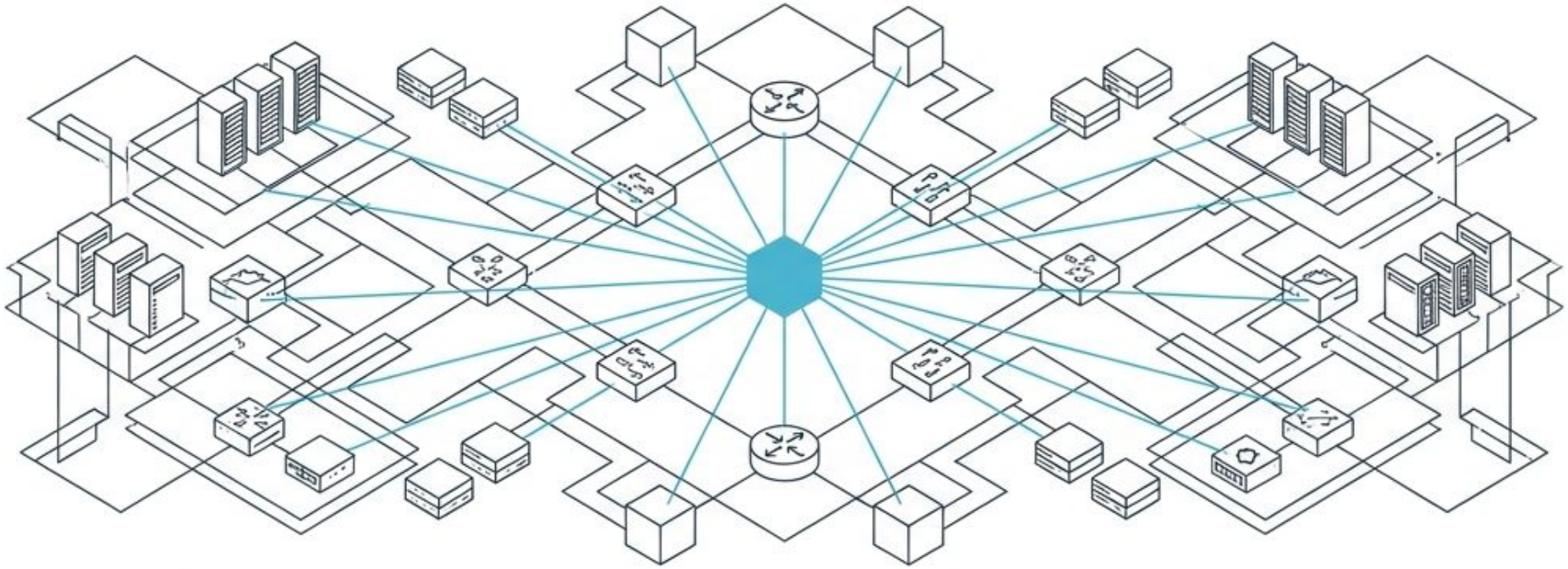
Beating the Three-Month Discovery Trap

A Strategic Blueprint for the Two-Week VMware Exit Assessment



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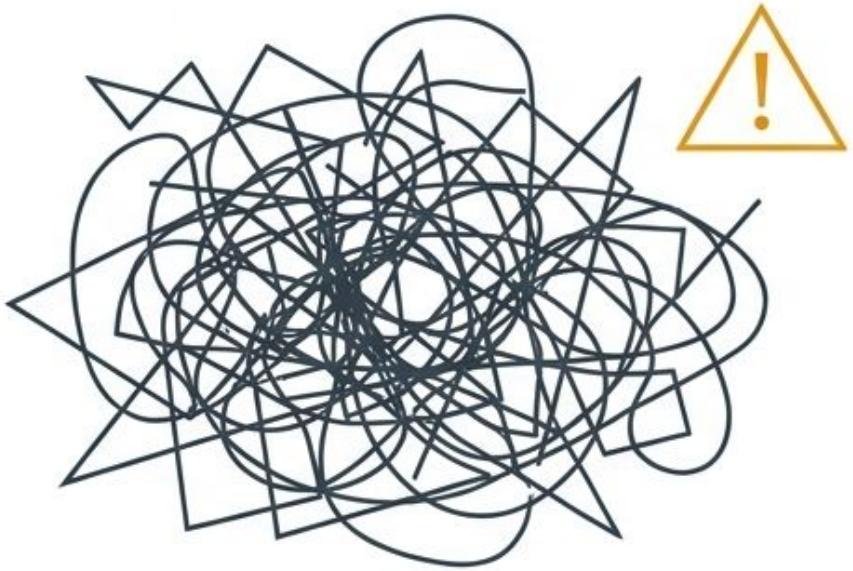
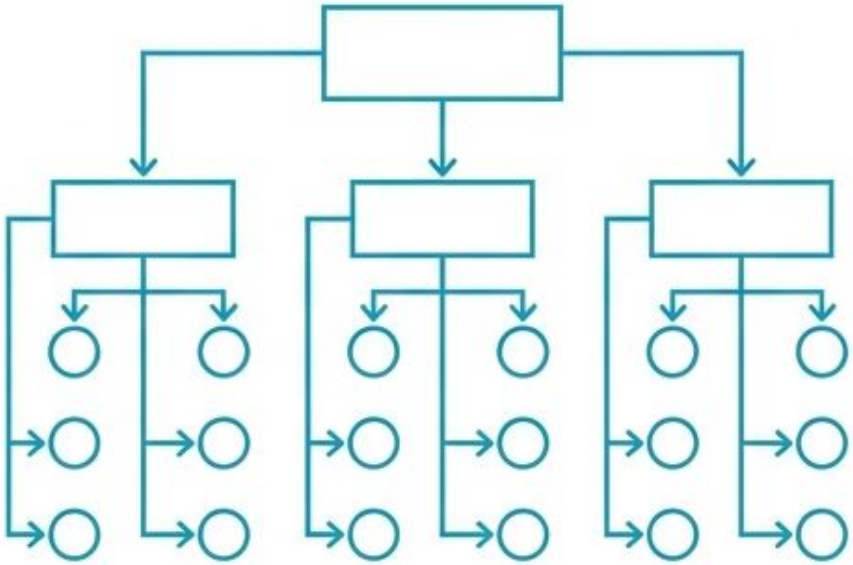
The Migration Discovery Bottleneck

Manual data collection and spreadsheet reconciliation do not scale. They introduce critical errors and delay the high-level cost frameworks required for executive sign-off.



THE REALITY: Gartner Baseline: 7-10 FTEs for 1 Month + 9 Months Evaluation



THE GOAL: The 14-Day Imperative: Inventory, Scope, Risk, Cost

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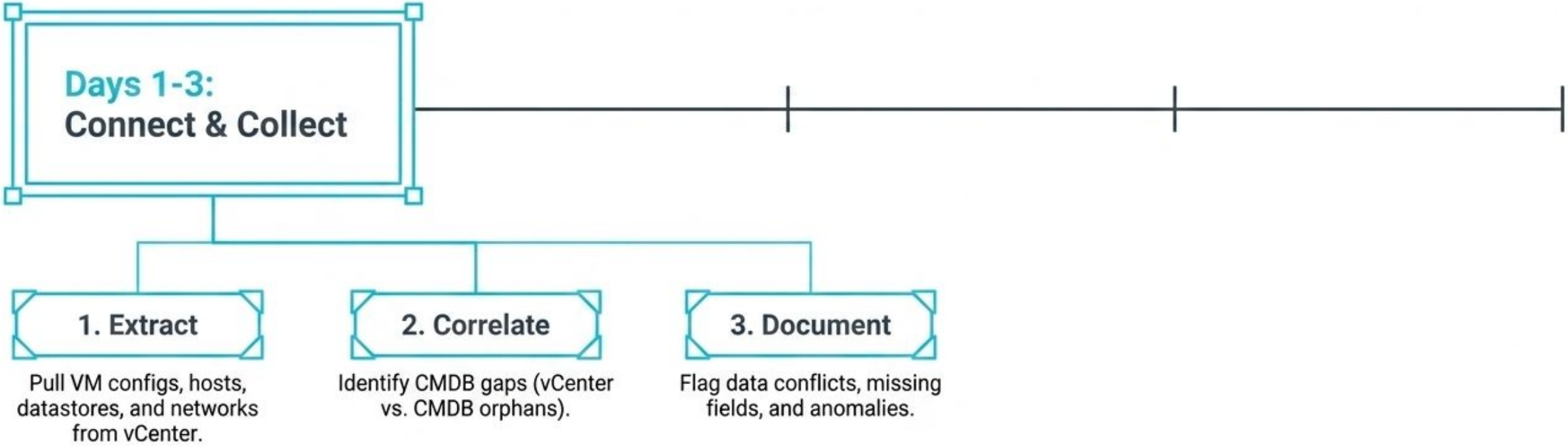
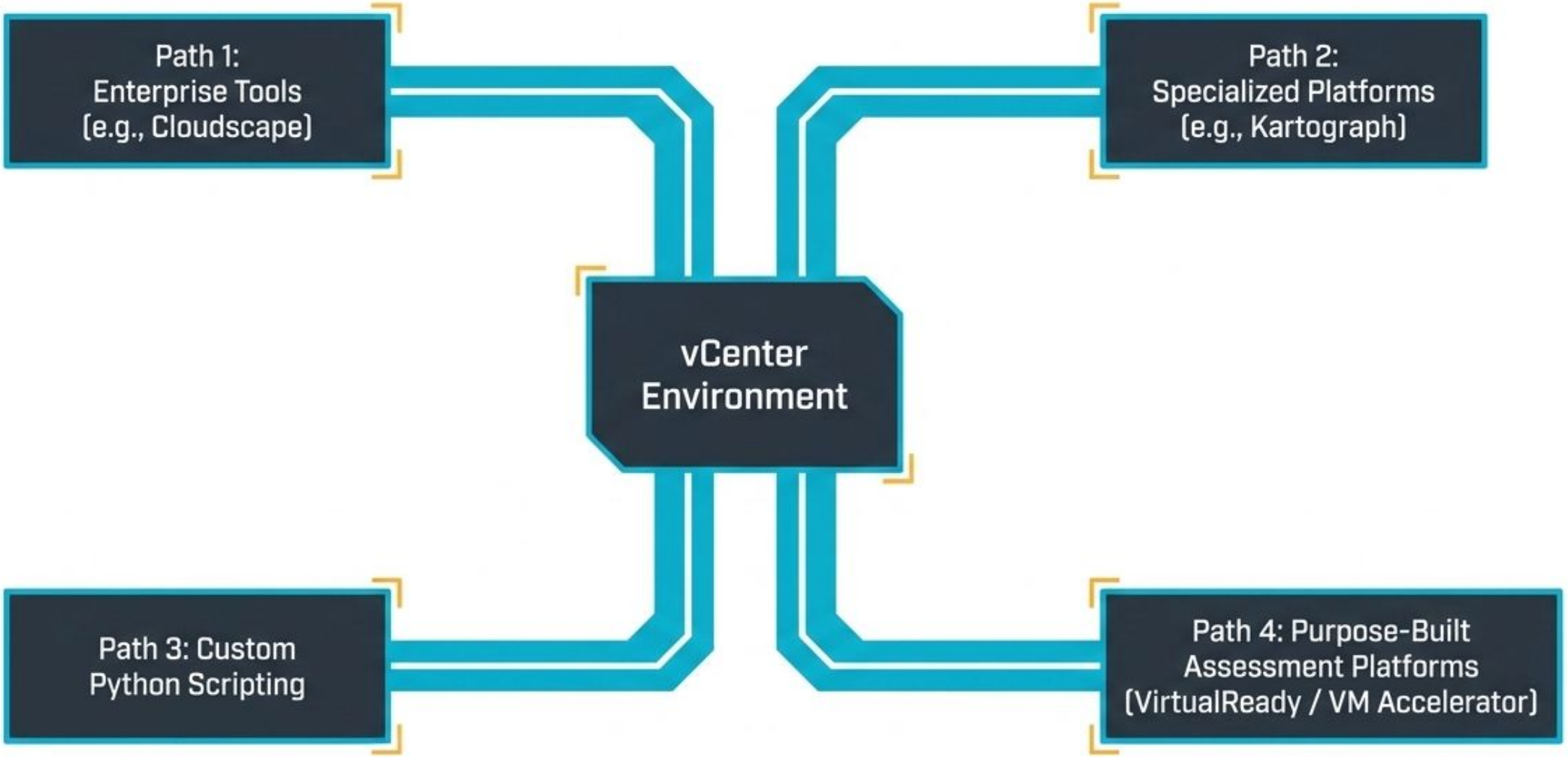
The Rapid Assessment Framework



Four Paths to Automated vCenter Discovery



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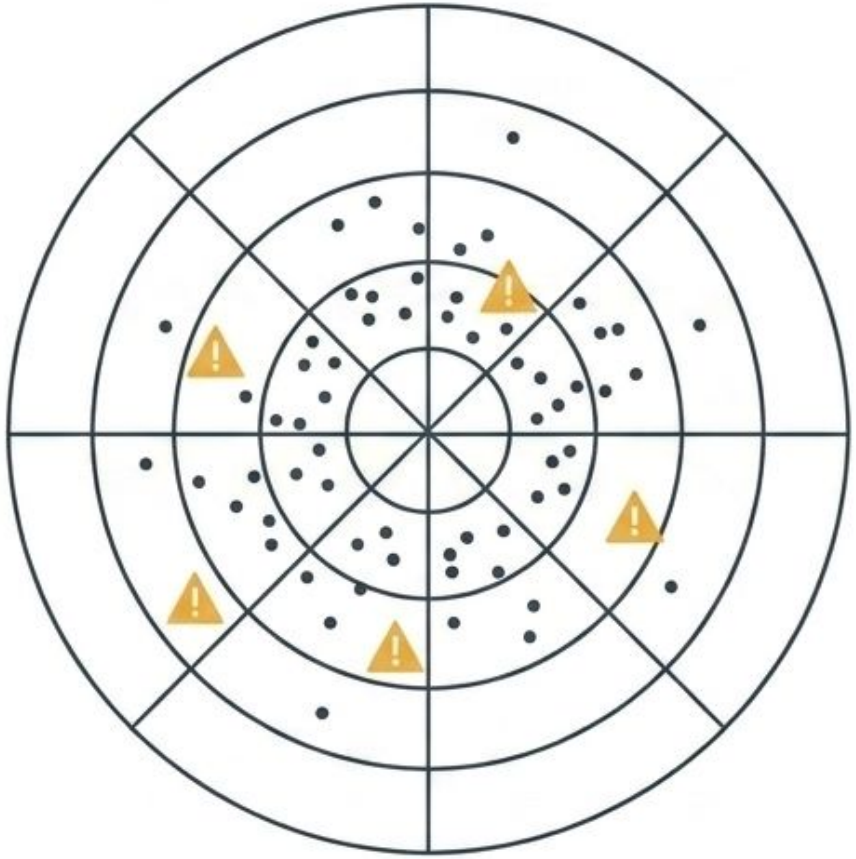
Path 1: Enterprise Tools (Cloudscape)



Typical Cost
"Free" (Often baked into existing cloud spend).

Key Advantage
Good for a quick baseline VM inventory and basic TCO modeling.

The Catch
Produces massive amounts of noisy data. Lack of context requires significant, time-consuming manual cleanup by your engineers.

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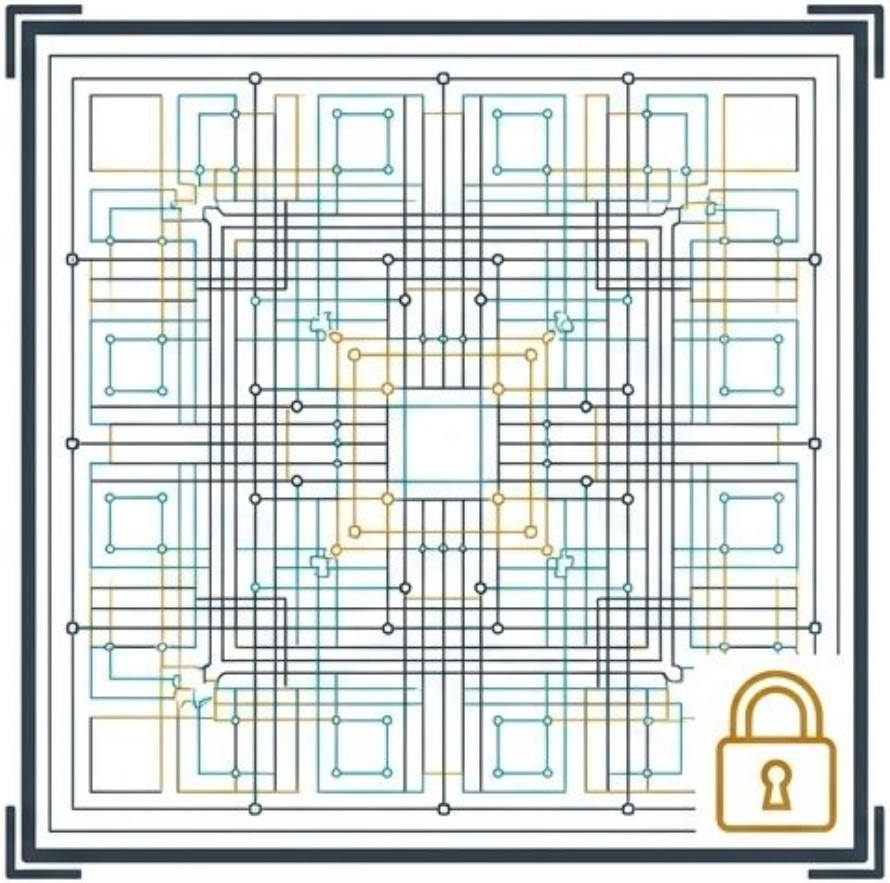
Path 2: Specialized Platforms (Kartograph)



Typical Cost
$50,000 to $150,000+ in licensing.

Key Advantage
Deep, application-level analysis. Vastly more accurate than native tools.

The Catch
Exorbitant licensing fees and heavy, complex upfront configuration that threatens the 14-day timeline.

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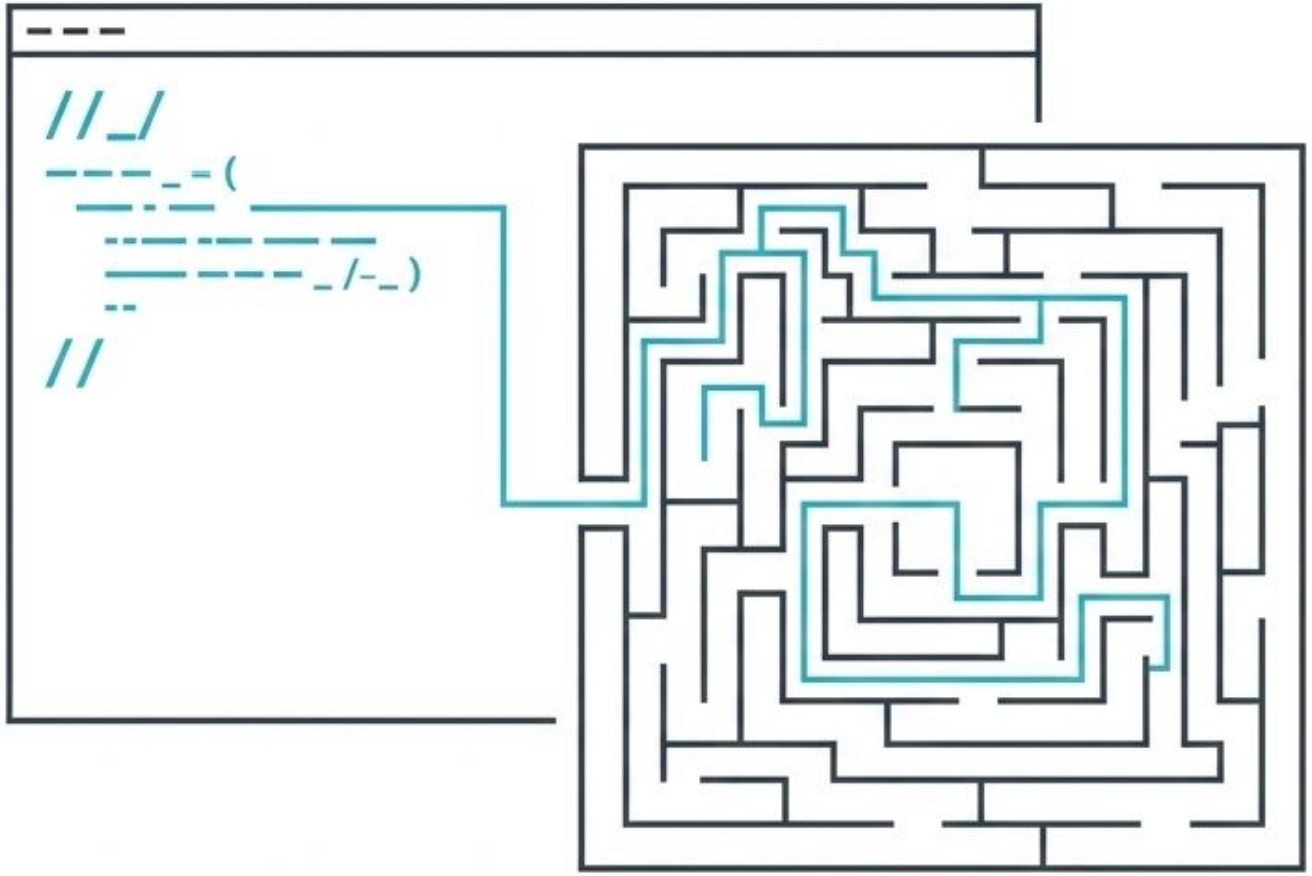
Path 3: Custom Python Scripting



Typical Cost

~$200 in raw engineering time.

Key Advantage

Delivers a hyper-precise, context-aware map by querying CMDBs, parsing netstat logs, and digging through config files.

The Catch

Demands elite engineering bandwidth. Your smartest engineers must know exactly what to look for and where.

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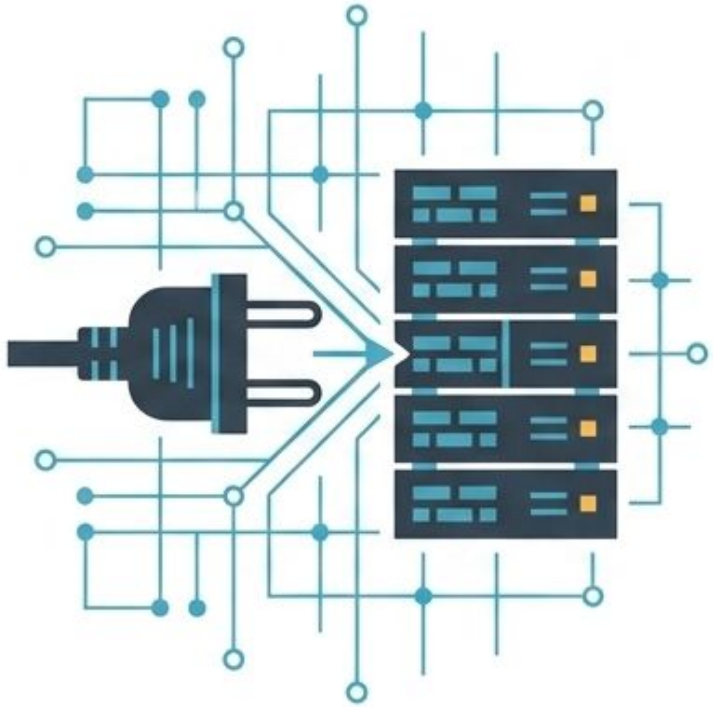
Path 4: Purpose-Built Assessment Platforms



Tool Profile
ReadyWorks VM Accelerator / VirtualReady

Key Advantage
The "Day-One" Connector. Installs in hours, connects to vCenter immediately, and provides initial inventory by the end of Day 1.

The Difference
Time-boxed delivery. Specifically designed for a rapid 45-day access window to output Risk Registers and complete discovery without perpetual procurement cycles.

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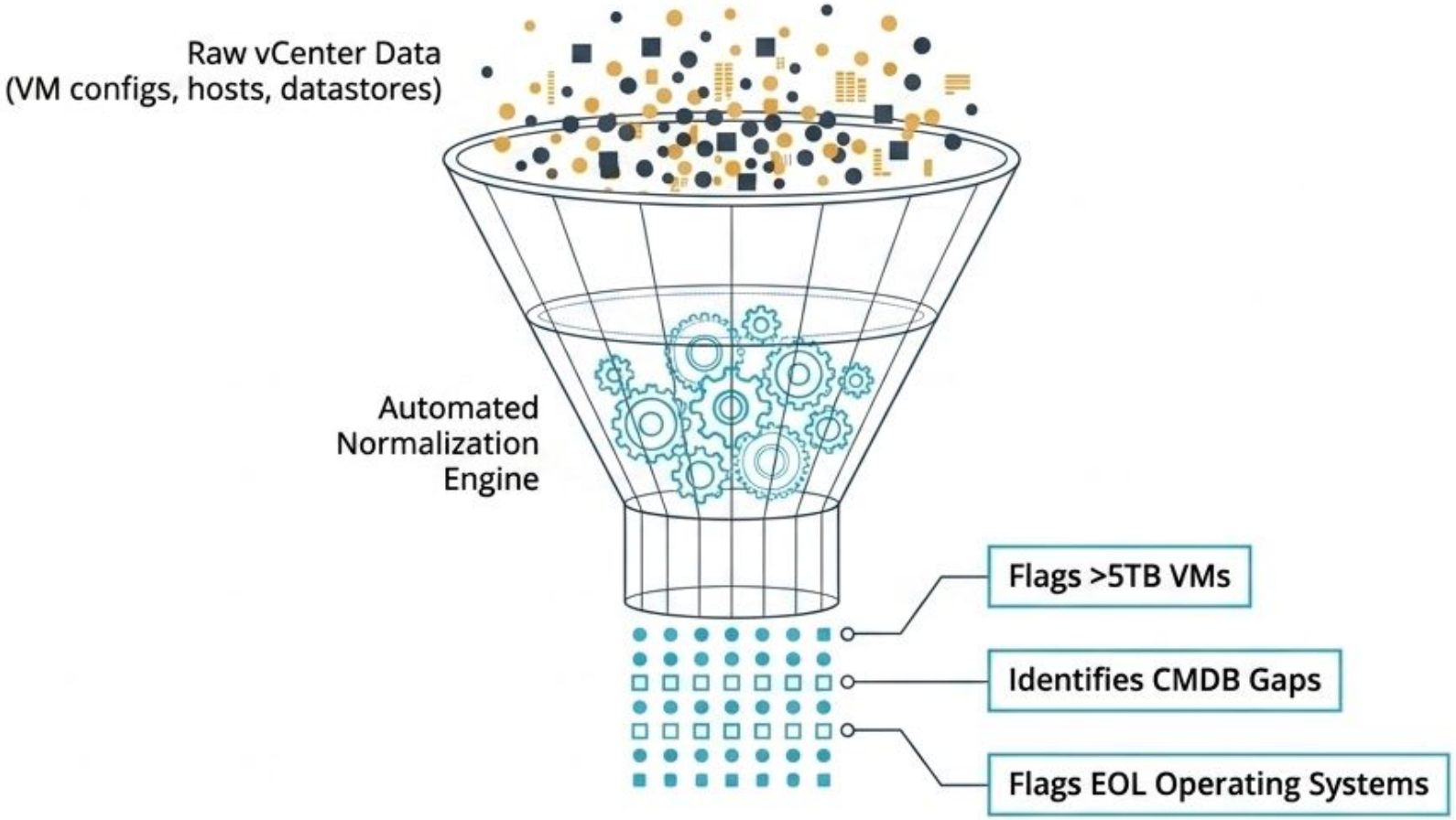
Beating the Clock via Automated Normalization



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The vCenter Discovery Automation Matrix

Automation Path	Typical Cost	Primary Advantage	Primary Limitation	Data Quality
Enterprise Tools	'Free'	Quick TCO modeling	High manual cleanup	Raw & Noisy
Specialized Platforms	$50k-$150k	Deep application analysis	Expensive / Heavy config	Deep & Complex
Custom Python Scripting	~$200 dev time	Highly precise / custom	Drains elite engineering	Context-Aware
Purpose-Built (VM Accelerator)	Time-boxed	Day-One vCenter connect	Assessment-focused	Auto-Normalized

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Flashy Dashboards vs. Real Assessments

A word of warning: Do not mistake a tool's flashy dashboard for a real assessment. Automated discovery is just step one. The real work is translating raw data into non-negotiable metrics that dictate migration feasibility.

Reference: Modernization Intel: Cloud Migration Risk Assessment

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The 14-Day Deliverables

Not a finished migration plan, but the definitive intelligence required to make informed Go/No-Go decisions.

1. Detailed Inventory

Complete, normalized list of VMs, compute, storage, OS, and ownership.

2. Risk Register

Preliminary view of blockers (EOL systems, >5TB storage, VMware-specific dependencies).

3. Cost & Timeline Framework

Reliable VM counts by category for accurate TCO modeling.

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The Blueprint for Rapid Discovery

Automate, Do Not Manually Scrape

Manual efforts guarantee failure on tight timelines.

Protect the Scope

Defer requests for deeper analysis to the execution phase. Use stakeholders for validation, not discovery.

Deploy Purpose-Built Tooling

Utilize platforms like ReadyWorks VM Accelerator for immediate, zero-friction vCenter ingestion.

Two weeks of discovery followed by a clear decision is better than three months of analysis followed by lingering uncertainty.

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Thank you

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